

Report: Assignment 1 (DA3408)

Question 1: Technical Debt Diagnosis

1) Part 1

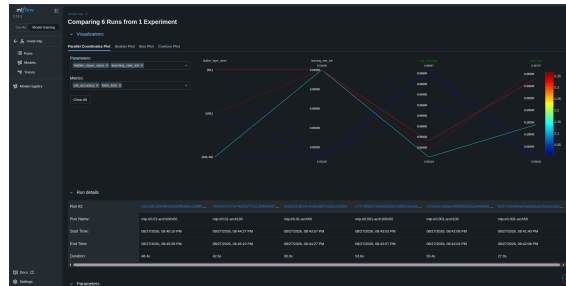
- a) Entanglement (CACE)
- b) Undeclared Consumers
- c) Configuration and Glue-Code Debt

2) Mitigating Problem (a): changing the “estimated delivery time” feature’s rounding logic unexpectedly hurt recommendations for a completely unrelated “favorite restaurants” feature:

I propose that the startup uses **MLflow Experiments** tracking to compare how incremental changes to the baseline model affects the model metrics. In this manner, they can pin the exact run, i.e., the exact model that caused a change in an unrelated feature (recommendations or favorite restaurants, etc.). To do so, they need to ensure that with every run of the experiment, they log the necessary metrics and the model artifacts properly.

Question 2: MLflow Experiment Comparison

1) MLflow UI



2) Analysis

The parameters varied for the experiment are learning rate (0.01, 0.001) and the MLP architecture ((100,), (100, 50), (50,)). The best performing run is **m1p-lr0.001-arch100** (MLP with learning rate 0.001, one hidden layer with 100 units) with the highest **test accuracy 0.9704** and **test F1-macro 0.9701**. **lr = 0.001** gives the Adam optimizer sufficiently small steps to converge in 25 steps with final **train loss 0.007141**, and the number of units = 100 is sufficiently large to model the digit classes without overfitting.

There is evidence of some overfitting when comparing train loss and test accuracy for **lr = 0.001**: despite 2 hidden layers, the (100, 50) architecture (hidden layer 1 has 100 units, hidden layer 2 has 50 units) has a **higher training loss (0.01303)** and **lower test accuracy (0.966)** when compared to the (100,) model (**loss: 0.007141, accuracy: 0.9704**). This is evidence of overfitting, even though the accuracy itself is very high (0.966). However, for **lr = 0.01**, the train loss is high with both architectures because training has not yet converged, which is why it gives rise to lower test accuracies than **lr = 0.001**.

Learning rate (lr) is the hyperparameter that appears to have a larger effect on performance: for architecture (100,), **lr = 0.001**, **test_accuracy: 0.970429**; **lr = 0.01**, **test_accuracy = 0.953571**.

3) Script

You can find the exact script at `question_2/question_2.ipynb` in the repository.

Question 3: DVC Data Versioning & Rollback: Checkout Demonstration

```
(.venv) ananya-kishore@ananya:~/da3408-assignment1$ cd question_3/
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ wc -l dataset.csv
2801 dataset.csv
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ git checkout v1.0
Note: switching to 'v1.0'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

git switch -c <new-branch-name>

Or undo this operation with:

git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 210fef4 added dataset v1.0, 1800 images, 1801 lines
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ dvc checkout
Building workspace index [1.00 [00:00, 238entry/s]
Comparing indexes [2.00 [00:00, 2.42kentry/s]
Applying changes [1.00 [00:00, 240file/s]
dataset.csv
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ wc -l dataset.csv
1801 dataset.csv
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ git checkout main
Previous HEAD position was 210fef4 added dataset v1.0, 1800 images, 1801 lines
switched to branch 'main'
Your branch is up to date with 'origin/main'.
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ dvc checkout
Building workspace index [1.00 [00:00, 181entry/s]
Comparing indexes [2.00 [00:00, 3.32kentry/s]
Applying changes [1.00 [00:00, 216file/s]
dataset.csv
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$ dvc status
Data and pipelines are up to date.
(.venv) ananya-kishore@ananya:~/da3408-assignment1/question_3$
```